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Implementation of a wireless sensor network for irrigation management in drip irrigation systems

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Water scarcity and inefficient irrigation practices pose significant challenges to modern agriculture, particularly in semi-arid regions. Traditional irrigation methods often lead to excessive water consumption and uneven water distribution, reducing overall efficiency and sustainability. Effective water management in irrigated agriculture is crucial for maintaining high crop productivity and alleviating water scarcity. To improve water management techniques, the integration of drip irrigation systems with wireless sensor networks (WSNs) was investigated herein. In the proposed system, valve control and soil moisture monitoring nodes were added to obtain real-time data on irrigation water volume (IWV), flowrate, pipe inlet pressure, and soil moisture within the WSN. The hardware and software designs of the soil moisture monitoring and valve control nodes were created, including the selection of components, specific functionalities of software algorithms, and communication protocols used for data transmission. A field test was performed in an olive orchard to assess the performance of the proposed system during the irrigation season. The results showed that soil moisture content varied between 42.0% and 24.4%, 44.6% and 25.9%, and 46.7% and 30.2% at different depths, validating the system's capability to optimize irrigation scheduling. The system's automated valve control capability enabled the precise application of irrigation water volume of 2,209 m^3 throughout the irrigation season. Flowrate stabilization at approximately 2.8 m^3/h and real-time pressure monitoring enabled early detection of anomalies such as clogging or low water supply, enhancing irrigation efficiency. The results of this study highlight the potential of WSNs in improving water management practices for irrigation. By providing real-time data and remote-control capabilities, the integrated system offers a tool for optimizing water usage and conserving water resources.

Keywords Wireless sensor network, Drip irrigation system, Soil moisture monitoring, Flow rate monitoring, Pressure monitoring, Olive tree

Abbreviations

5G	Fifth generation wireless network
ADC	Analog to digital converter
AES-CTR	Advanced encryption standard in counter mode
EEPROM	Electrically erasable programmable read-only memory
ET	Evapotranspiration
FC	Field capacity of soil which indicates the moisture retained in the soil against gravity
GPIO	General purpose input/output
IoT	Internet of things
JSON	JavaScript object notation
LoRa	Long range wireless communication protocol
LTE-M	Long-term evolution for machines
MCU	Microcontroller unit
MQTT	Minimum Message Queuing Telemetry Transport
NB-IoT	Narrow band internet of things
RF	Radio frequency
RF-Rx	Radio frequency reception

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RF-Tx	Radio frequency transmission
TCP/IP	Transmission control protocol/internet protocol
USART	Universal synchronous/asynchronous receiver-transmitter
VWC	Volumetric water content of soil
WSN	Wireless sensor network

Water is the major resource for achieving and retaining high crop yields in irrigated agriculture. The increasing use of water for industrial, domestic, and recreational purposes[1] and climate change have led to water shortages[2,3]. Almost 70% of freshwater resources are used in agriculture, mainly for irrigation[4,5]. Water, despite being a limited resource that must be efficiently used, it is overused in agriculture. This causes salinization, particularly in semiarid areas depending on the water quality[1,6].

Although irrigation plays a crucial role in ensuring food security by increasing agricultural productivity, traditional irrigation methods have notable drawbacks, including inefficient use of water, fertilizer, and energy resources. Inefficiencies in these conventional approaches frequently result from inadequate monitoring, reliance on farmer experience rather than data-driven decision-making, and absence of real-time control capabilities. Therefore, the increasing pressures of climate change and water scarcity underscore the urgent necessity of developing sustainable, precise, and adaptive irrigation techniques capable of optimizing resource use, maintaining agricultural productivity, and reducing the environmental footprint of farming practices.

Drip irrigation is deemed valuable as it increases crop yield; promotes efficient use of water, fertilizers, and energy; and can be automated[7–9]. In this method, water supplied from a water resource is delivered to the irrigated area via a pressurized pipe system and applied to the root zone of plants. Frequent water supply in small amounts improves the soil moisture conditions during crop development; therefore, it is key to sustaining high crop yield and quality[10]. Despite the advantages of drip irrigation in agricultural water management, operating the system like conventional irrigation systems—characterized by manual operation, reliance of farmer experience and lack of real-time monitoring of critical parameters such as soil moisture, pressure, flowrate and water volume often lead to over- or under-irrigation, negatively affecting crop yield, quality, resource conservation and decision-making processes.

Irrigation management involves decision-making on the application time and amount of water to crop root zones throughout the irrigation season[11]. Among irrigation scheduling methods[12–14] such as time-based operation, soil moisture monitoring provides essential and precise inputs into decision support and irrigation automation systems used for irrigation management[15–19]. A predefined soil moisture setpoint, which indicates readily available water content in the corresponding soil layer, is usually measured as volumetric water content (VWC, %Pv) or soil water tension (kPa) between field capacity and wilting point. This soil moisture setpoint was also used to generate start condition for irrigation events[13,20,21]. This trigger point varies based on the soil and plant type, water resource capacity, and environmental factors. Contrastingly, the stop condition can be i) a value representing a soil moisture higher than the moisture at the start condition, ii) irrigation water volume (IWV) if a water meter is available, or iii) duration of irrigation if no water meter is available and the number and flowrate of drippers are known.

A drip irrigation system must have optimum hydraulic features, such as suitable operating pressure, for its successful operation. Low pressure levels can cause emitter clogging and result in the uneven distribution of water and fertilizer across the field due to decreased coefficient of variation, causing inefficient fertigation. High pressure may also cause physical damage and leaks in the main, submain, and lateral pipelines[22–24]. Therefore, information obtained from pressure sensors installed on pipes at valve points, i.e., plot heads, can avert and mitigate such problems.

Electronic systems and sensors are currently used for irrigation management; however, irrigation management has rapidly evolved over the years due to the emergence of license-free low power communication frequencies and protocols[25], ultra-low power microcontrollers[26,27], miniaturization, and innovative sensor designs[28,29]. In addition to advancements in hardware design of irrigation systems, manufacturers and community members have facilitated and accelerated the development of related software. This has fostered dynamic and rapid progress in embedded firmware and frontend/backend design of irrigation systems. Owing to these developments, more than 25.4 billion devices will be connected to the internet in 2030, with 152,200 devices per minute, which will generate 73.1 zeta bytes of data by 2025[30].

Wireless sensor networks (WSNs) are a prominent outcome of such developments; they can transmit data obtained from physical phenomena through sensors via radio signals within certain limitations[28,31]. WSNs are currently used in military, health, environment, infrastructure, and agriculture industries for monitoring and control[32].

Olive grove areas have reached 10.949 million ha in 2022 globally. As it demands special climate, 10.194 million ha (93%) of this total area is located in 10 Mediterranean countries, primarily in Spain, Tunisia, Italy, Morocco, and Turkey[33]. Olive tree is tolerant to water stress because its root system adapts to water-scarce conditions[34,35]; however, its yield and quality increases or decreases based on the irrigation strategies employed[36,37]. However, the application of WSNs in olive orchards has not been intensively reported. An exception is the LoRaWAN-based irrigation system deployed in a 22-hectare olive grove in Greece by Liopa-Tsakalidi et al., which demonstrated the benefits of multi-depth soil moisture monitoring, leading to a 42% reduction in water consumption in 2020 and 25% in 2021[38].

In recent years, several studies have explored the implementation and effectiveness of IoT-based frameworks in optimizing agricultural resource utilization. For instance, Jeličić et al. and Zervopoulos et al. presented the hardware, software, and field tests of WSN nodes developed for environmental and soil moisture monitoring using ZigBee compliant transceivers, respectively[39,40]. Sitharthan et al. developed an AI-driven autonomous irrigation system utilizing 6G-enabled IoT networks, which integrates machine learning prediction models

with weather and soil moisture data, achieving 86.34% accuracy in controlled environments[41]. Galaverni et al. investigated the effect of different irrigation regimes on tomato plants using an IoT-driven data collection platform, successfully optimizing irrigation scheduling by monitoring chlorophyll content and fluorescence[42]. Furthermore, Et-taibi et al. proposed a cloud-integrated smart irrigation system to enhance water management across multiple farms, particularly in arid regions[43]. IoT-enabled irrigation has also been extensively studied for water conservation. Jamshidi et al. designed an IoT-driven system for apple orchards, focusing on disease prediction, irrigation, and fertilization optimization, leading to a 33% reduction in fungicide use and a 50% decrease in pesticide application[44]. In another application, Rose et al. introduced an IoT-enabled dendrometer system, which continuously monitors trunk diameter fluctuations in Christmas trees as an indicator of water stress and irrigation efficiency[45]. Further advancements include solar-powered and AI-enhanced smart irrigation solutions. Mamun et al. developed a solar-powered smart irrigation system that integrates ESP8266 microcontrollers integrated with capacitive soil moisture sensors and Minimum Message Queuing Telemetry Transport (MQTT) protocols for real-time soil moisture monitoring, successfully optimizing pump activation to reduce fossil fuel dependence[46]. Shahab et al. proposed a smart soil monitoring system that measures moisture, salinity, electrical conductivity, pH, nitrogen, phosphorus, and potassium levels in real time, offering AI-driven recommendations for irrigation and fertilization[47]. A recent study carried out in Tunisia demonstrated that deficit irrigation at 75% of crop evapotranspiration, managed via smart tensiometers, effectively maintained winter wheat yield while significantly improving water use efficiency, making it a viable strategy for precision irrigation in water-limited environments[48]. To address key challenges in smart irrigation, such as limited data sources, low automation, and reliance on cloud computing, a LoRa-based edge computing system was developed for precision irrigation. The system integrates STM32 MCUs, WH- 101-L low-power LoRa modules, 4G communication, and high-precision GPS, enabling real-time soil moisture and meteorological data collection. Wireless communication tests confirmed stable data transmission up to 4 km, with transmission rates varying from 0.9 s (10.937 kbps) to 6 s (0.268 kbps) for 100 bytes. Field trials during the wheat grain-filling stage demonstrated that the irrigation algorithm maintained soil moisture above 90% of the optimal range, ensuring efficient water management[49]. To demonstrate the application of IoT technologies, an IoT-driven framework including web, desktop and mobile applications, integrating sensor technology and AI-driven data analysis was implemented to improve decision making and real-time operational control in smart greenhouse prototype[50]. An IoT-based drip irrigation system was developed and tested for Eggplant to improve irrigation efficiency. The system utilized wireless sensors and a microcontroller to monitor relative humidity, soil and air temperature, and soil moisture content, with data transmitted to a cloud server for remote access. Compared to conventional crop evapotranspiration (ETc) -based drip irrigation, the IoT-based system resulted in 12.05% higher crop yield and 35.2% water saving, demonstrating its potential for optimizing irrigation scheduling and resource efficiency[51]. In a study conducted by Kumar et al., a LoRa-based smart agriculture management system operating at 868 MHz frequency was developed to enhance irrigation, crop management, and environmental monitoring. Wireless sensors were deployed to collect soil moisture, crop growth, and weather data, transmitting information to a central management system for analysis and decision-making[52]. In another study, an IoT-based irrigation system was developed to enhance efficiency and sustainability in agriculture. Solar-powered ESP32 microcontrollers and low-energy, long-range communication technologies were integrated to enable real-time data transmission from soil moisture, pH, temperature, humidity, and PIR sensors to a cloud server. The system was tested in the field, demonstrating its ability to optimize water use, reduce operational costs, and improve crop yields, particularly benefiting farmers in remote areas with limited infrastructure[53].

Although pressure measurement is a critical factor in maintaining optimal system performance, it remains as underexplored topic in the IoT-based research with limited studies addressing its role in irrigation. An example study utilizing an IoT-based fertigation system demonstrated that pressure measurements play a crucial role in determining irrigation system readiness for fertilization, with the pressure sensor confirming the operational status of the irrigation system[54]. A recent study has highlighted the potential of smart water distribution system to enhance water demand forecasting, leakage detection, quality monitoring, and agent-based distribution strategies, thereby improving overall energy efficiency and reducing operational costs in urban water management systems[55]. Karray et al. conducted an experimental and comparative study on a multi-layer IoT architecture designed to interact and cooperate in detecting and locating leaks in water pipelines[56]. Sadeghioon et al. developed and tested a smart wireless sensor network that utilizes indirect pressure measurements in plastic pipes to detect and locate leaks, demonstrating its effectiveness through both laboratory and field experiments[57]. In a study conducted in Indonesia a hardware and software for a real-time water pressure monitoring was successfully tested for 5 min data interval with nearly 100% data retrieval[58].

Despite significant advancements in IoT-based smart irrigation and WSNs, most studies primarily focus on soil moisture monitoring without critical parameters such as pipe pressure and flowrate and relied on commercial development boards like Arduino, Raspberry Pi, or ESP32. In this study design, implementation, and field validation of a WSN based irrigation monitoring and control system with custom-developed hardware, firmware and communication protocol, specifically for precision agriculture applications, with a particular focus on olive orchards, is aimed. Study integrates real-time measurement and control of multiple critical irrigation parameters, including soil moisture content, pressure, flowrate and total water volume, and customized configuration structure to manage the system's behavior, thereby ensuring more precise and sustainable irrigation management. Additionally, this study combines the monitoring of critical irrigation parameters with automated valve control software and hardware architecture, effectively addressing the common drawbacks of operating irrigation systems relying solely on farmer experience. Proposed system's modular hardware architecture, structured in main and submain layers, allows for future enhancements, such as integration of soil electrical conductivity and temperature sensors and enables easy integration of various wireless communication protocols such as LoRa and NB-IoT, making it a scalable solution for advanced irrigation management.

The structure of this paper is organized as follows: In the 1st section system architecture, including the WSN design and integration with the drip irrigation system is described. Hardware and software implementations of soil moisture monitoring and valve control nodes are introduced in the 2nd section. Field test setup and methodology is presented in the 3rd section, while the experimental results obtained from the olive orchard trial are discussed in the 4 th section. Finally, key findings, highlighting limitations, and suggesting directions for future research are concluded in the 5th section.

System architecture

The system architecture components, besides the WSN, are not the subject of this study; however, they are briefly mentioned herein. The system architecture comprised a WSN domain with a drip irrigation system, a cloud domain containing a dedicated Linux server running web application(s) and MySQL database, and a user domain wherein web and mobile client applications were serviced (Fig. 1).

Data transmission between the WSN and the server was established by the 3G Modem (Telit, Model HE910) connected to a WSN coordinator (Tier 1) over RS- 232 and to cloud over Netsocket with Transmission Control Protocol/Internet Protocol (TCP/IP). It was programmed using Python 2.7 and could be bidirectionally operated. A Linux machine connected to MySQL database was used on the cloud as Infrastructure as a Service. JavaScript and PHP programming languages were used for developing web applications for remote monitoring and valve control. Some additional libraries such as jQuery, e-charts 3.0, and Material Design Bootstrap were used for data visualization, document object model (DOM) manipulation, and event handling.

Wireless sensor network

Transceiver selection

LE70-868 short-range radio modules (Telit) operating in the license-free ISM frequency band of 868 MHz were selected at the physical layer of the network for all nodes, including the coordinator. The maximum theoretical range and transmitting power of the transceivers were 10 km and 27 dBm (0.5 W), respectively.

WSN topology

The Telit star network with smart repeater functionality was used as the WSN topology as its network structure was suitable for the structure of the drip irrigation system. A combination of star and line networks was used as the operating mode, restricted to three tiers (Fig. 2). In this topology, the central node, i.e., the coordinator, could communicate with the nodes outside its direct radio coverage via intermediate nodes (repeaters) in the network[59].

The maximum number of nodes and sub-nodes in the network was determined based on the following rules[59]:

- The maximum branch number is 16.
- The maximum sub-branch number (i.e., sub-node number) under any node is 15.
- The maximum node number on a branch is 255.

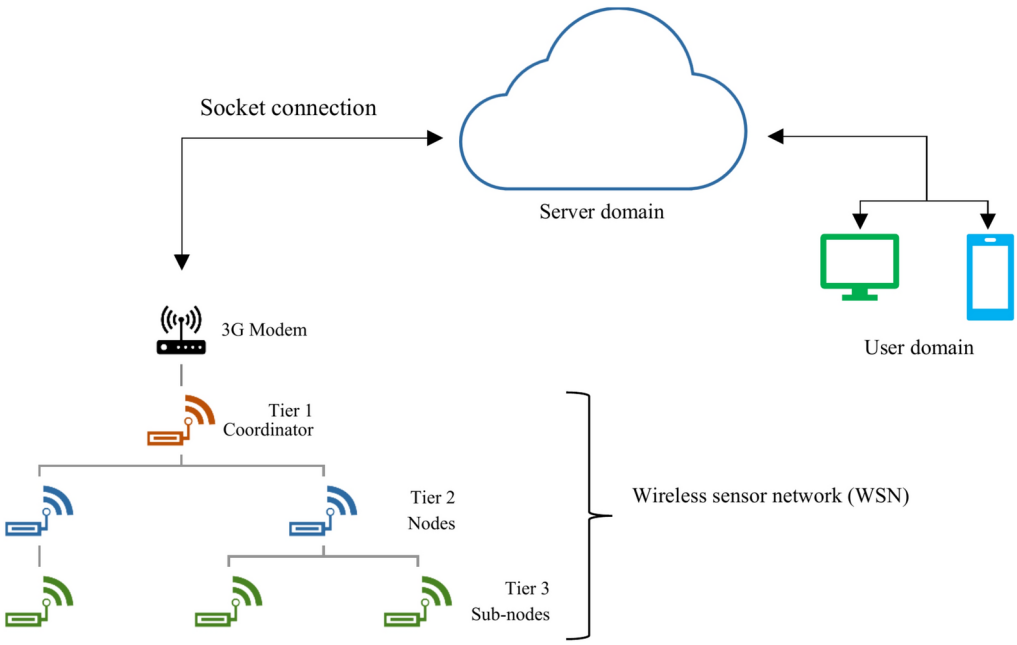


Fig. 1. General architecture of the proposed system.

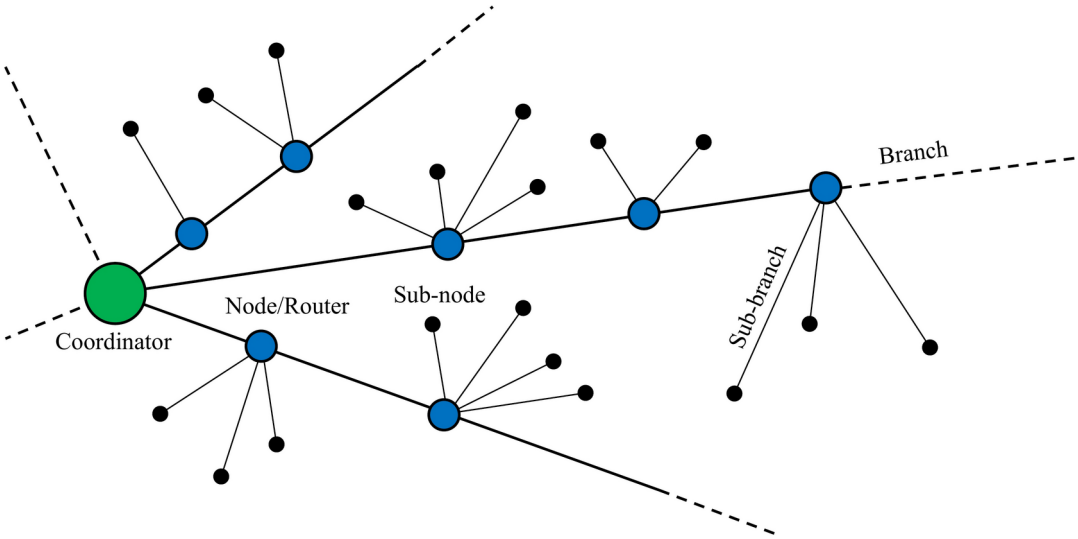


Fig. 2. Topology of a wireless sensor network.

	Soil monitoring node	Valve control node
Coordinator	–	+
Node	–	+
Sub-node	+	–
Router	–	+

Table 1. Network node–drip irrigation components/node matrix.

The coordinator was the main access point of the network, over which all the data between server and WSN were transmitted. The following rules were applied for communication within the WSN[59]:

- The coordinator can send and receive data from any node or sub-node in a network.
- The coordinator can send data to all the sub-nodes under any node.
- The coordinator can send data to all the nodes and sub-nodes over any branch.
- Any node can act as a repeater to increase the radio cover range.
- A node cannot send and receive messages from any other node.
- A sub-node cannot send and receive messages from any other sub-node.

Overlaying WSN topology with the drip irrigation system

Before overlaying, the drip irrigation system nodes were considered as soil moisture monitoring and valve control nodes. To determine the node or sub-node of the WSN topology that could represent the drip irrigation system node(s), a network node–drip irrigation node matrix was created (Table 1). According to the matrix, soil moisture monitoring can be represented only by a sub-node, whereas valve control can be represented only by a node. As sub-nodes could not act as a repeater due to WSN communication rules, only the valve node acted as the repeater while performing its specific tasks for the drip irrigation system. In contrast, soil moisture monitoring node could only be placed under a valve node. In this case, this node and its sub-nodes virtually represented the monitoring and control of a plot in an agricultural enterprise.

Hardware design

The soil moisture and valve control nodes of the drip irrigation system (Fig. 3) comprise main and submain hardware layers (Fig. 4); however, only the radio module, required electronics, and serial connection (RS- 232) required for establishing communication with the GSM module were included in the coordinator.

Similar main hardware layers were designed for soil moisture and valve control nodes; they included the STM32 F401RE 32-bit ARM Cortex-M4 microcontroller (MCU) (ST Microelectronics), LE70 - 868 short-range radio module, AT24C512 two-wire serial electronically erasable programmable read-only memory (eeprom, Atmel), and other electronic components such as resistors and capacitors. It could be plugged to the submain layer using row headers. The power domain along with all analog and digital inputs/outputs designed for the specific requirements of the sensor node were placed on the submain hardware layer.

MCU handled sensor reading, basic calculations, preparing Javascript Object Notation (JSON) formatted data, sending data to the radio module over the universal synchronous/asynchronous receiver/transmitter (USART) transmit line (Tx), the evaluation of node configuration parameters received from the server via the

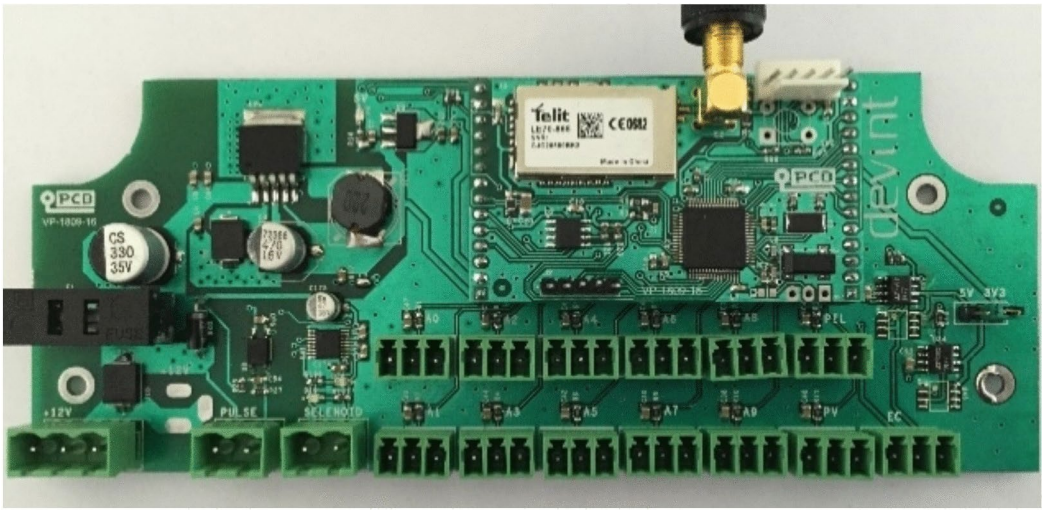


Fig. 3. Soil moisture monitoring and valve control nodes of a drip irrigation system with the same hardware design.

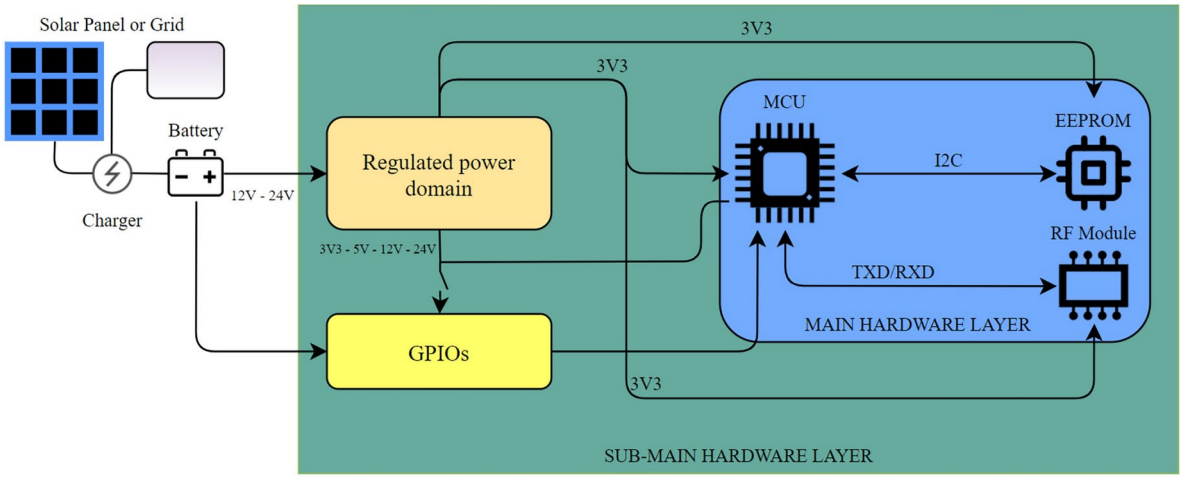


Fig. 4. Block diagram of the drip irrigation wireless sensor node.

USART receive line (Rx), reading and writing operations on eeprom for parameter saving and loading, and monitoring battery voltage. When powered up, the MCU configured nodes according to the default parameters and continued with its normal operation mode. In case of an external or internal interrupt, it jumped and completed the interrupt service routine (ISR) and returned to the main software loop. The source of the interrupt could be USART-Rx connected to a radio module, an internal timer for data transmission, a digital input signal from a water meter, and an alarm condition such as low or high pressures based on sensor reading, among others.

The radio module established communication between the sensor node and coordinator. The analog and digital inputs/outputs of the radio module were disabled, except USART, as the general-purpose inputs and outputs (GPIOs) of the MCU were used. Both the radio and USART baud rate of the module were 38,400 bps, and radio Rx was always on. However, radio transmission (RF-Tx) only occurred when an interrupt was generated in the MCU relating to data transmission or alarm condition or when any response was given to a specific request or query sent by the server. The submain hardware layer was designed according to specific power and sensor requirements of drip irrigation system nodes. However, the same design was applied to both soil moisture monitoring and valve control nodes. Tables 2 and 3 show the properties and the supported sensor and device characteristics of this layer, respectively. For example, the latching solenoid valve was controlled by the DRV8800 DC Motor Driver (Texas Instruments, USA) enabled with an H-bridge functionality.

The soil moisture monitoring and valve control nodes in drip irrigation systems are distributed over an agricultural field; therefore, energy cabling is not viable. In the proposed system, the software and hardware of the nodes were designed to operate on a solar charged gel/lead acid battery. Thus, all sensors were powered down between measurements using a dedicated GPIO (Fig. 4). Tables 4 and 5 present the current consumption of soil

	Soil monitoring node	Valve node
Supply voltage	12 VDC	12 VDC
Regulated voltage output to main hardware layer	3.3 VDC	3.3 VDC
Regulated voltage output to GPIOs	3.3 VDC, 5 VDC	3.3 VDC, 12 VDC
Number of analog inputs	12 (9)*	2
Number of digital inputs	–	1
Number of digital outputs	–	1
Number of digital input/outputs as SDI- 12 line	1	–
Number of RS485 lines	–	–

Table 2. Properties of the submain hardware layer. *Three of them reserved for future use.

	Soil monitoring node	Valve control node
Sensor/Device[1]	Soil moisture sensor	Pressure sensor
Supply voltage	3.3 or 5 VDC	3.3 VDC
Output	0–3.3 VDC	0–3.3 VDC
Sensor/Device[2]	Soil moisture sensor (SDI- 12)	Water meter with reed sensor
Supply voltage	5 VDC	-
Output	Digital	Pulse
Sensor/Device	Battery monitoring	Latching solenoid valve[3]
Supply voltage	–	9 - 12 VDC
Output	0–14 VDC to 0–3.3 VDC	–
Sensor/Device	–	Battery monitoring
Supply voltage	–	–
Output	–	0–14 VDC scaled to 0–3.3 VDC

Table 3. Supported sensor and device characteristics of the submain hardware layer. [1]Such as Teros 10 or ECH_2O 10HS (Meter Group). [2]Such as Teros 12 (Meter Group). [3]Controlled by the DRV8800 DC Motor Driver (Texas Instruments) enabled with H-bridge functionality.

	Quantity	Current per component (mAh)	Operation interval per component (minutes/sensor)	Total operation number per component (time/hour)	Operation duration per component (milliseconds/hour)	Total current consumption (mAh)
Analog soil moisture sensors	12	15	15	4	40[1]	0.002
SDI- 12 Digital sensors	3	30	15	4	600[2]	0.015
RF-Tx	1	335	15	4	120[3]	0.011
RF-Rx	1	25	–	–	3,600,000	25
MCU	1	7	–	–	3,600,000	7
TOTAL	–	–	-–	–	–	**32.0**

Table 4. Current consumption of the soil monitoring node. [1]Measurement duration of a soil moisture sensor is considered as 10 ms. [2]Measurement duration of an SDI- 12 digital sensor is considered as 150 ms. [3]Approximate transmission duration of 125 bytes of data at 38,400 bps 8 N1 is considered as 30 ms.

moisture monitoring and valve control nodes based on these modifications, respectively. Both nodes consume 32 mAh of current. The radio module with an always-on Rx state and MCU consume almost 100% of current. Sensors, valves, and radio module with a Tx state consume negligible current because of their low measurement/transmission duty cycle, low power requirement of the sensors, and short operation duration.

Software design

The MCU firmware, assembled in the main hardware layer of each drip irrigation system node, was programmed with C/C++ using STM32 CubeIDE (ST Microelectronics). Due to their operational characteristics and differences, each node had individual embedded software (Figs. 5 and 6).

Both nodes in the main loop of the embedded firmware had similar initialization procedures. When the nodes were powered up, USART, Inter-Integrated Circuit (I2C), and a software timer were initialized for radio, eeprom communication, and data transmission timing, respectively, in addition to hardware abstraction library peripheral initialization.

	Quantity	Current per component (mAh)	Operation interval per component (minutes/sensor)	Total operation number per component (time/hour)	Operation duration per component (milliseconds/hour)	Total current consumption (mAh)
Analog pressure sensors	2	2.1	0.5	120	720[1]	0.000
Latching solenoid valve[4]	1	350	1,440	0.08	0.833[2]	0.000
RF-Tx	1	335	15	4	120[3]	0.011
RF-Rx	1	25	–	-	3,600,000	25
MCU	1	7	–	-	3,600,000	7
TOTAL	–	–	–	-	–	**32.0**

Table 5. Current consumption of the valve node. [1]Measurement duration of an analog pressure sensor is considered as 6 ms. [2]Operation duration of each phase (ON and OFF) is considered as 10 ms, separately. [3]Approximate transmission duration of 125 bytes of data at 38,400 bps 8 N1 is considered as 30 ms. [4]Operation frequency is considered once a day as On/Off.

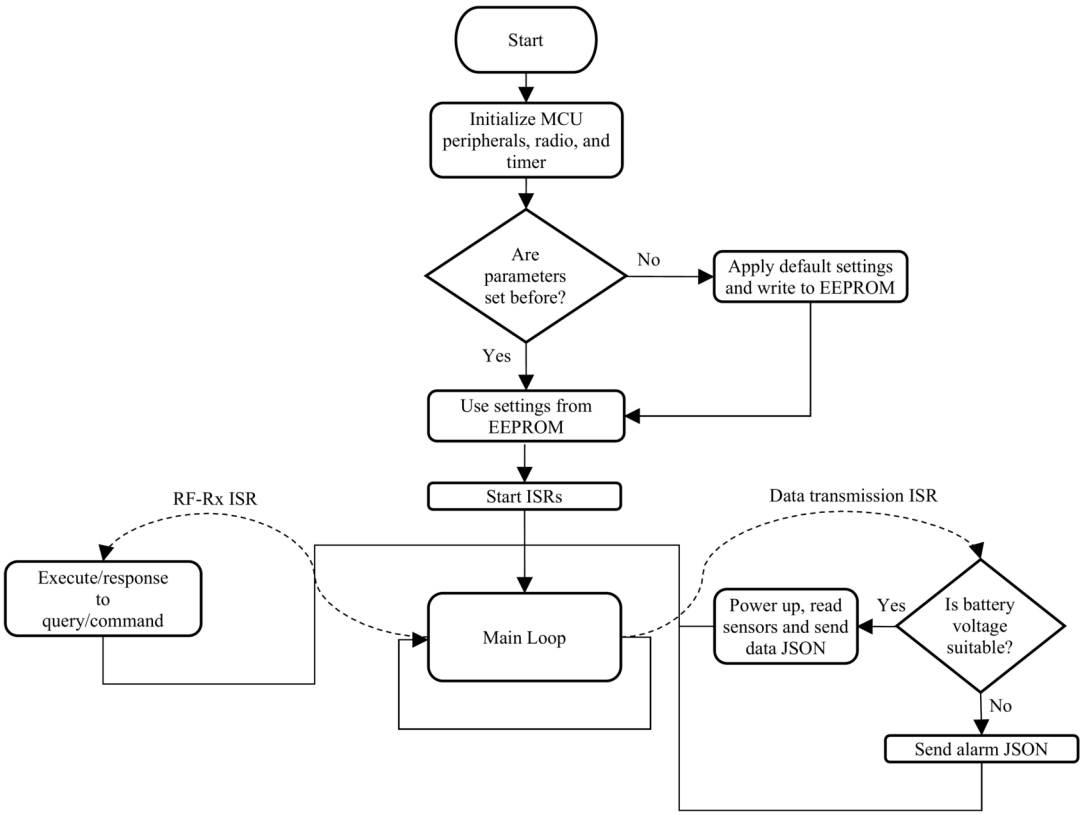


Fig. 5. Software flowchart of the soil moisture monitoring node.

After initialization, the MCU checked the specific address of eeprom for a specific value to determine whether this node was set before the main loop based on the parameters listed in Table 6. If it was the first operation of the node, the MCU wrote predefined default values to eeprom and used them. Otherwise, it directly read eeprom for the mentioned parameters. The operational parameters were set and get on runtime with "NSETSYS" and "NGETCFG" for soil moisture node and "SSETSYS" and "SGETCFG" for valve node commands (Table 7). After parameter setting, the interrupt service routines were attached to USART-Rx and the timer. The remaining process was interrupt driven and managed by data transmission and USART-Rx ISRs (Fig. 7).

Soil monitoring node

In the soil moisture monitoring node, data transmission triggering interrupted the MCU. The MCU then checked the battery voltage for correct soil moisture sensor readings that were determined as a minimum of 9.5 VDC, corresponding to 2,210 mV or 2,742 as 12-bit analog to digital converter (ADC) value (Fig. 8).

If the battery voltage was suitable, i.e., >9.5 VDC, the MCU read soil moisture sensors, prepared JSON-formatted data (Table 8), sent them to the WSN coordinator, and returned to its main loop. Otherwise, it sent a JSON formatted alarm indicating that the battery voltage was unsuitable for soil moisture sensor measurement (Tables 8 and 9). As the maximum allowable data length for the radio module was ~255 bytes, JSON data

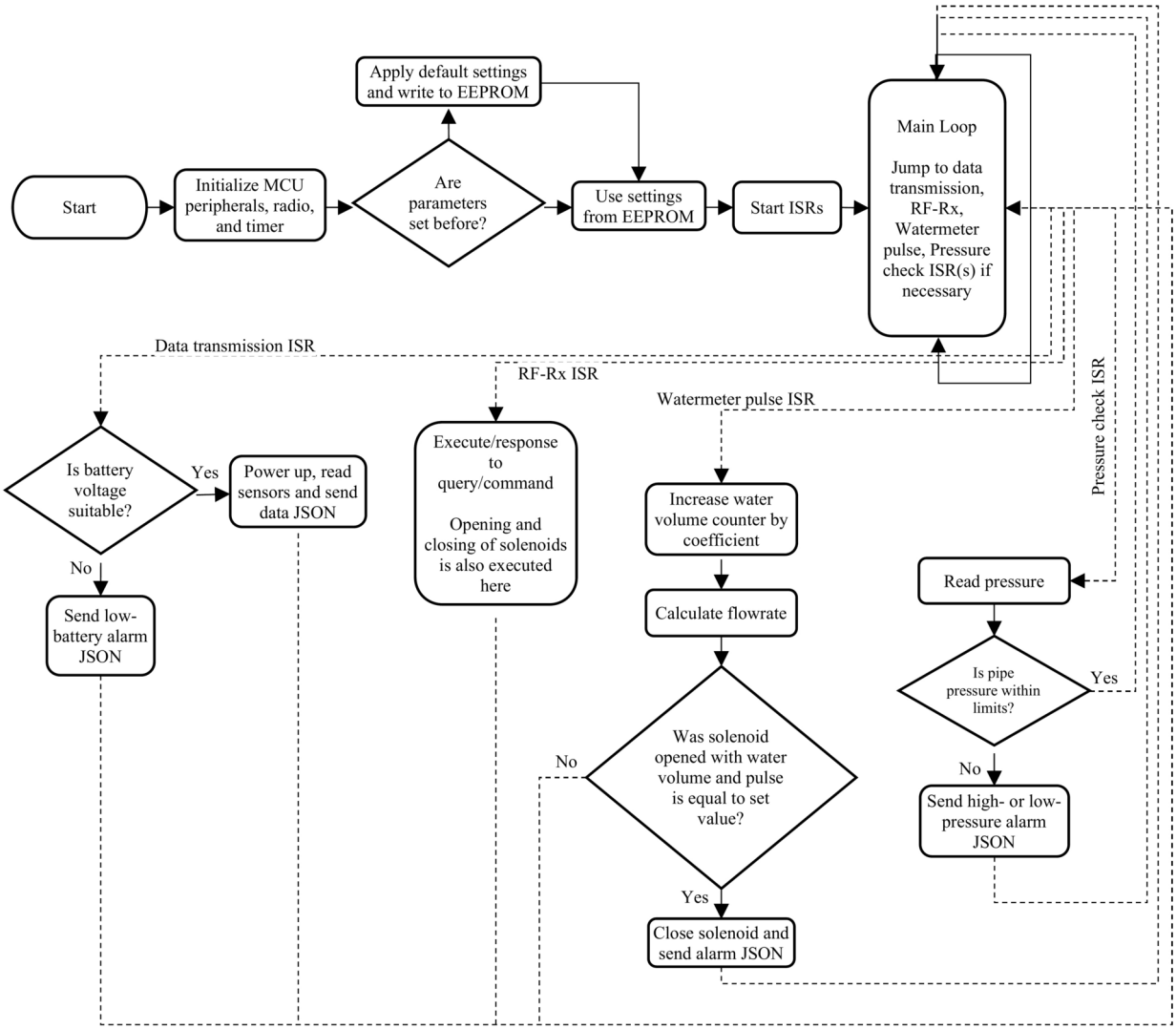


Fig. 6. Software flowchart of the valve control node.

structure length was limited to 200 bytes in each transmit by the MCU firmware. Additionally, if any serial digital interface (SDI-12) soil sensor was connected to a node, three other transmissions including SDI-12 soil electrical conductivity (GSE), soil temperature (GST), and soil dielectric (GSD) readings, occurred.

The first three bytes of each transmitted data were prepared by the radio module: it's the first two bytes indicate the node address, and the third byte was always 0x3D (`=`). The last byte of the JSON-formatted data was always 0x0D (carriage return, \r). The soil moisture sensor (except SDI-12 sensors) readings were transmitted as 12-bit ADC values between 0 and 4,095. Each ADC channel was sampled 16 times, and the final value was calculated using the root mean square method. The actual soil moisture readings were determined via server-side processing using the soil moisture calibration equation. Its coefficients could be defined by users in the server side. In contrast, 12-bit ADC battery voltage readings were determined in the node and as a server-side process using a regression equation (Fig. 8). The reason of node-side calculation is sensor reading decision mentioned above, while server-side calculation is data presentation. Similarly, RF-Rx interrupt triggering signaled the main loop. Then, the MCU checked whether the incoming data are known commands (such as MCU reset, set parameters again, immediately read sensors, and take test measurement), executed related code, and returned to the main loop.

Valve control node

In the valve node, the main software loop broke due to the triggering of data transmission interrupt. Then, the MCU checked the battery voltage, read pressure sensor(s), totalized water-meter pulses, calculated flowrate, and prepared JSON-formatted data (Table 8). In JSON-formatted data, the flowrate was calculated in liter/s by dividing the IWV between two data transmissions (liter) to data transmission interval (seconds). The units of irrigation water (WI), cumulative irrigation water (WIC), and flowrate (QI) were converted from liter, liter, and liter/s to m^3, m^3, and m^3/h respectively, on the server side for data visualization with the e-charts library.

	Soil monitoring node		Valve control node	
Set	**Value/parameter**	**Parameter no**	**Value/parameter**	**Parameter no**
Soil moisture sensor 1 connected?	0 or 1[a]	P1[b]	–	–
Soil moisture sensor 2 connected?	0 or 1[a]		–	
Soil moisture sensor 3 connected?	0 or 1[a]		–	
Soil moisture sensor 4 connected?	0 or 1[a]		–	
Soil moisture sensor 5 connected?	0 or 1[a]		–	
Soil moisture sensor 6 connected?	0 or 1[a]		–	
Soil moisture sensor 7 connected?	0 or 1[a]		–	
Soil moisture sensor 8 connected?	0 or 1[a]		–	
Soil moisture sensor 9 connected?	0 or 1[a]		–	
Battery voltage read enabled?	0 or 1[a]	P2	0 or 1[a]	P1
Any SDI- 12 sensor connected?	0 or 1[a]	P3	–	
Automatic data transmission enabled?	0 or 1[a]	P4	0 or 1[a]	P2
Automatic data transmission interval in the irrigation phase	Minute	P5	Minute	P3
Automatic data transmission interval in the non-irrigation phase	Minute	P6	Minute	P4
Pressure sensor connected?	–	–	0 or 1[a]	P5
Water-meter coefficient	–	–	1, 10 or 100 L/pulse	P6
Maximum pressure of pressure sensor	–	–	Bar	P7
Pressure unit	–	–	0: PSI, 1: Bar	P8
Pressure control interval	–	–	Second	P9
Pressure alarm low threshold	–	–	Bar	P10
Pressure alarm high threshold	–	–	Bar	P11
Pressure alarms enabled	–	–	0 or 1[a]	P12
Max allowed water volume after solenoid close	–	–	Liter	P13
Pressure threshold for non-irrigation phase	–	–	Bar	P14

Table 6. Configuration setting parameters of the soil monitoring and valve control nodes. [a] 0: Not connected or disabled, 1: Connected or enabled. [b] Stored as 16-bit unsigned integer value in soil monitoring node.

	Command	Usage	Description
Valve control node	SSETSYS	SSETSYS P1 … P14	Space separated configuration set
	SGETCFG	SGETCFG	Space separated configuration get
	SOPEN	SOPEN	Turn the solenoid valve on
	SWOPEN	SWOPEN <liter water>	Turn the solenoid valve on for indicated total liter of water. And then automatically turn off when total volume is reached
	SCLOSE	SCLOSE	Turn the solenoid valve off
Soil monitoring node	NSETSYS	NSETSYS P1 … P6	Space separated configuration set
	NGETCFG	NGETCFG	Space separated configuration get
	NIRR	NIRR	Switch data transmission interval to irrigation phase
	NDRY	NDRY	Switch data transmission interval to non-irrigation phase

Table 7. Configuration and solenoid valve and irrigation phase control commands in the valve and soil monitoring nodes.

An additional interrupt routine, i.e., reading the pressure sensor installed on a pipe at the plot entrance to check inlet (operating) pressure during or after irrigation, helped to monitor uniform water distribution across the plot. Three pressure evaluation regions, i.e., low, normal, and high, were defined based on the low and high thresholds of the pressure alarm (Table 6 and Fig. 9). A phase change triggered the JSON-formatted alarm, which indicated whether the inlet pressure was in the safe pressure region (Tables 8 and 9). The commands sent by the server are used to control the solenoid valve in the valve node (Table 7). If the valve node received "SOPEN," the solenoid valve was turned on until the "SCLOSE" command was received. Contrastingly, if the command "SWOPEN <liter water> " was received, the valve node turned the solenoid valve on and automatically turned it off after the desired volume of water passed through the water meter indicated by " < liter water >." At the end of each process, a specific alarm with code 6 or 7 was sent (Table 9). Each on and off cycle changed the value of "STAT" key to "IRR" or "DRY" in the JSON data shown in Table 8.

The data transmission interval in both nodes was changed when the solenoid valve was turned on or off in the valve node. This transition in the soil moisture node was enabled by the "NIRR" and "NDRY" commands sent

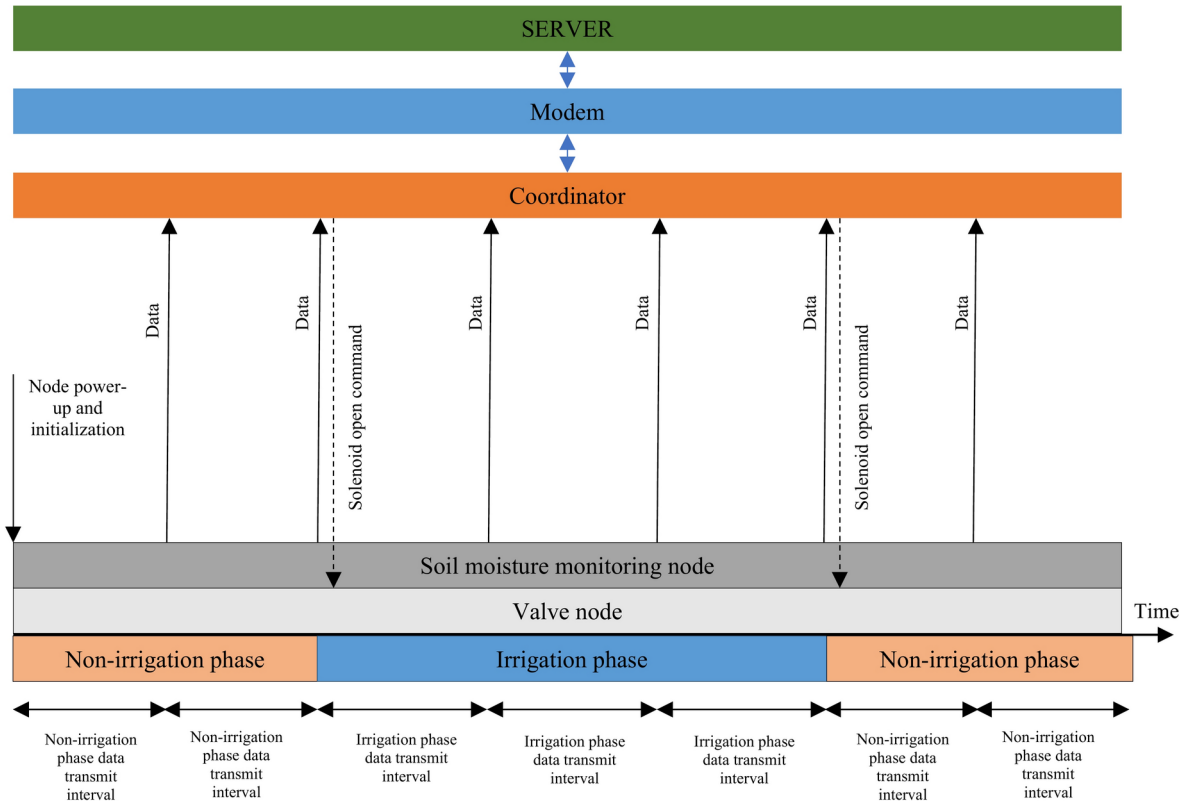


Fig. 7. Interrupt-driven data transmission between the node and server.

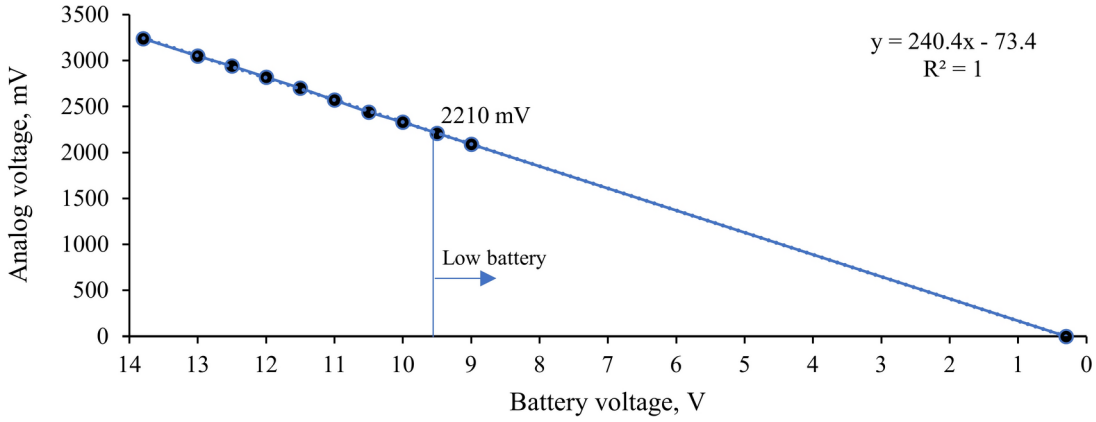


Fig. 8. Relation between analog and battery voltages.

by the server side (Table 7). As the soil moisture content increased rapidly during irrigation than that during the non-irrigation period, such independent dual transmission intervals enabled better monitoring of fluctuations in soil water content during irrigation.

Soil moisture sensor

In this study, the Teros 10 and Teros 12 soil moisture sensors (Meter Group, USA), were integrated into the WSN-based irrigation system to ensure accurate and real-time soil moisture monitoring. While both sensors were incorporated into the system architecture, field tests were conducted specifically using the Teros 10.

The Teros 10 is a capacitive soil moisture sensor designed for VWC measurement. It operates at a dielectric measurement frequency of 70 MHz, which minimizes sensitivity to soil salinity (EC). The sensor provides a VWC measurement range of 0–0.64 m^3/m^3 with a resolution of 0.001 m^3/m^3 and an accuracy of ± 0.03 m^3/m^3in mineral soils with a volume of influence of 430 ml and solution EC < 8 dS/m. The sensor supports supply voltage

		1 st byte of node id	2nd byte of node id	3rd byte	JSON	Last byte
Soil monitoring node	Data	0x00 - 0xFF	0x00 - 0xFF	0x3D	{N:{A0:..,A1:..,A2:..,A3:..,A4:..,A5:..,A6:..,A7:..,A8:..,B:…,„STAT:"…"}}[1]	0x0D
					{GSE:{E1:..,E2:..,E3:..,E4:..,E5:..,E6:..,E7:..,E8:..,E9:..}}[2]	
					{GST:{T1:..,T2:..,T3:..,T4:..,T5:..,T6:..,T7:..,T8:..,T9:..}}[3]	
					{GSD:{D1:..,D2:..,D3:..,D4:..,D5:..,D6:..,D7:..,D8:..,D9:..}}[4]	
	Alarm	0x00 - 0xFF	0x00 - 0xFF	0x3D	{ALR:"NALR",ALRCODE:…,B:…}[5]	0x0D
Valve control node	Data	0x00 - 0xFF	0x00 - 0xFF	0x3D	{S:{PR:..,WI:..,WIC:..,QI:..,B:..,STAT:"…"}}[6]	0x0D
	Alarm	0x00 - 0xFF	0x00 - 0xFF	0x3D	{ALR:"SALR",ALRCODE:…,B:…}[7]	0x0D

Table 8. Data and alarm structure of the soil monitoring and valve nodes. [1]N: Indicates that data were sent by the soil moisture node, A0–A9: 12-bit ADC readings of nine soil moisture sensors, B: 12-bit ADC reading of the battery voltage, STAT: If the value of STAT key is IRR or DRY node is in the irrigation or non-irrigation phase, which directly affects the data transmission interval, respectively. [2]E: Electrical conductivity reading from SDI-12 sensor, μmhos.cm^{-1}. [3]T: Soil temperature reading from SDI-12 sensor, °C. [4]E: Dielectric reading from SDI-12 sensor, unitless. [5]NALR: Indicates that alarm was sent by the soil moisture node, ALRCODE: Soil moisture node specific alarm code (Table 8), B: 12-bit ADC reading of battery voltage. [6]S: Indicates that data were sent by the valve node, PR: Pressure sensor reading (bar), WI: Water volume between two data transmission (liter), WIC: Cumulative water volume between the start and end of irrigation (liter), QI: Flow rate between two data transmission (l.s^{-1}), B: 12-bit ADC reading of the battery voltage, STAT: If the value of STAT key is IRR or DRY node is in the irrigation or non-irrigation phase, which directly affects the data transmission interval, respectively. [7]SALR: Indicates that alarm was sent by the valve node, ALRCODE: Valve node specific alarm code (Table 8), B: 12-bit ADC reading of the battery voltage.

	Alarm Code	Alarm Description
Soil monitoring node	1	Low battery voltage (< 9.5 VDC). Sensor readings will be disabled
	2	Normal battery voltage (> 9.5 VDC). Sensor readings will be enabled
Valve control node	1	Low pressure during irrigation
	2	Pressure is in the safe region during irrigation
	3	High pressure during irrigation
	4	Low pressure after irrigation
	5	High pressure after irrigation
	6	Solenoid valve turned on. Irrigation started
	7	Solenoid valve turned off. Irrigation ended
	8	Low battery voltage (< 9.5 VDC). Solenoid valve will be turned off
	9*	Normal battery voltage (> 9.5 VDC)

Table 9. Generated alarms and alarm codes by the soil monitoring and valve control nodes. *Normal battery voltage (> 9.5 VDC) does not turn the solenoid valve on again if the turn off reason was low battery alarm (< 9.5 VDC).

range from 3.0 to 15.0 VDC, ensuring compatibility with various low-power IoT-based monitoring systems. Its measurement duration is 10 ms. The sensor provides an analog measurement output between 1,000–2,500 mV[60].

Field test

Materials and methods

The proposed WSN was tested in a 10-year-old olive orchard (82 trees) at the Olive Research Institute, Bornova-İzmir, Turkey (38°27`10.45``N, 27°11`56.90``E) during the 2019 irrigation season (from May 19 to October 15, 2019). The Olive Research Institute has a role as a specialized center for olive research in the region and represents semi-arid Mediterranean climate characteristics. Moreover, institute has readily available irrigation system and electricity infrastructure (water resource, pumps, main distribution pipes, cabling etc.).

Irrigation water was supplied from a nearby well with a submersible pump using a drip irrigation system comprising six drippers per tree (2 L/h each, in-line pressure compensated type). For the automatic operation of the pump during irrigation events, a hydrophore was installed on the main pipeline. Before the irrigation season, a soil monitoring node at a distance of 2 m from a selected tree, a valve control node at a point where main irrigation pipe connected with manifold line, and a coordinator connected to the 3G Modem were installed in the olive orchard (Fig. 10).

Irrigations were managed based on the measurements from six analog soil moisture sensors (Teros 10, Meter Group) connected to the soil moisture node; two soil moisture sensors were installed at a depth of 300 mm from

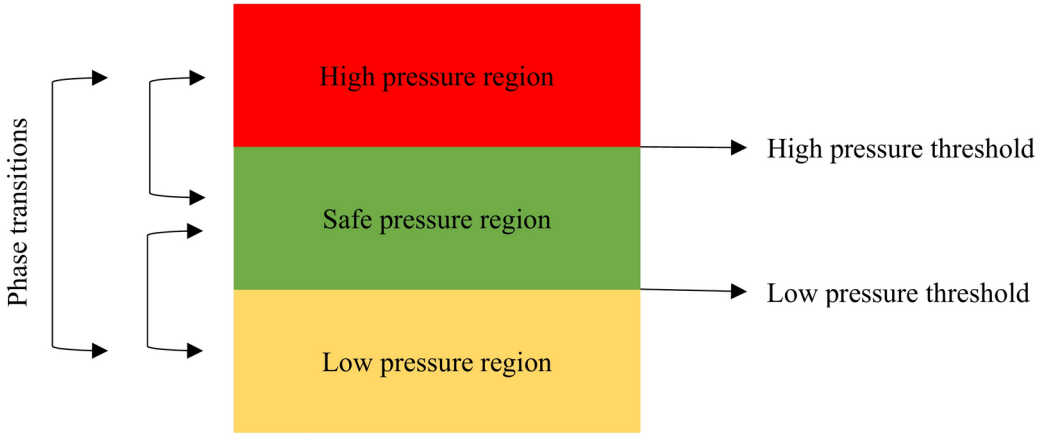


Fig. 9. Phase transitions between various pressure regions.

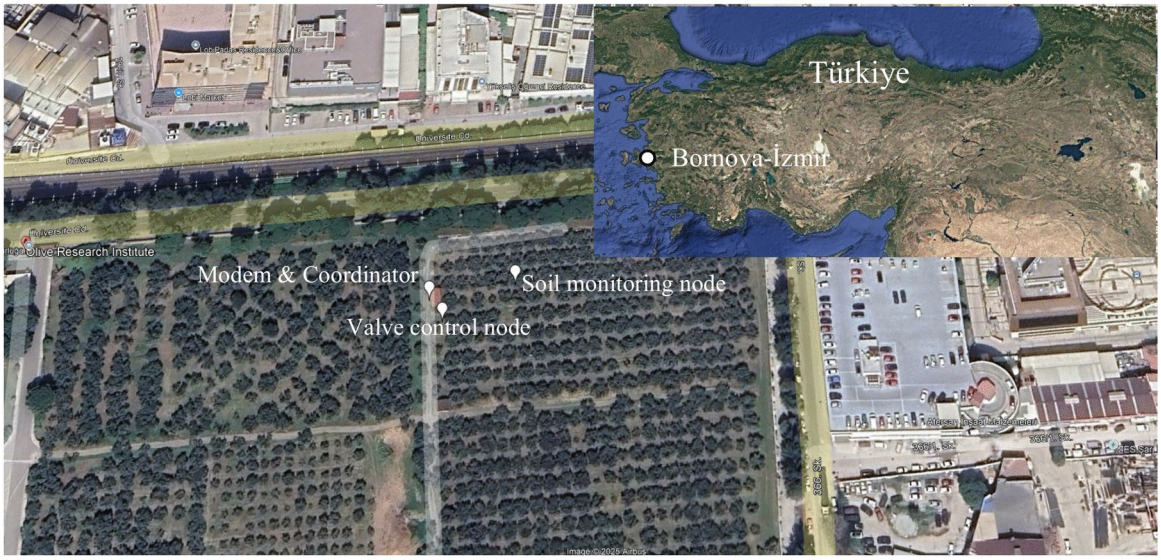


Fig. 10. Installation location of coordinator and modem and soil monitoring and valve control nodes in an olive orchard.

the soil layer (0–300, 300–600, and 600–900 mm). Soil moisture measurement intervals for the irrigation and non-irrigation phases were 5 and 15 min, respectively. A 6-bar 3.3 VDC pressure sensor (SSCSHNT060PGAA3, Honeywell), a pulse water meter (TK26, 1 L/pulse, Baylan), and a 9 VDC latching solenoid valve (150-PGA- 9 V, Rainbird) with a PRS-dial pressure regulator accessor (Rainbird) were connected to the valve control node to measure the inlet pressure and IWV and control irrigation (valve on/off) and manually adjust inlet pressure, respectively. The pressure control interval, pressure alarm low threshold, and pressure alarm high threshold were 30 s, 0.6 bar, and 2.5 bar, respectively. The inlet pressure was adjusted to 2 bar manually using the PRS-dial pressure regulator at the start of the irrigation season.

During the hardware and software development phase, the pressure sensors were verified within a range of 0–5 bar using a water pump connected to a closed pipe system. The verification was performed by comparing the measured values, calculated using Eq. 1, with those obtained from reference pressure sensors and analog manometers.

$$P_o = \frac{0.8 \cdot V_s}{P_{max} - P_{min}} \cdot (P_a - P_{min}) + 0.1 \cdot V_s \quad (1)$$

where.

P_o = Output of pressure sensor, V.
V_s = Supply voltage of pressure sensor, V.
P_{max} = Maximum pressure of pressure sensor within compensated pressure range, bar.
P_{min} = Minimum pressure of pressure sensor within compensated pressure range, bar.

P_a = Applied (measured) pressure, bar.

Additionally, pulse watermeter readings were validated throughout the irrigation season by comparing with mechanical readings of the watermeter, confirming their accuracy. On the other hand, the calibration of the Teros- 10 soil moisture sensors was performed using Eq. 2, as recommended by Meter Group. Operating at a high measurement frequency of 70 MHz, the sensor is less sensitive to variations in soil texture and electrical conductivity, and its generic calibration equation has been reported to provide sufficient accuracy under saturation extract conditions with electrical conductivity values up to 8 dS/m, yielding an absolute accuracy of ± 0.03 m^3/m^3in most mineral soils[60].

$$\%Pv = \left(\left(4.824 \cdot 10^{-10} \cdot mV^3\right) - \left(2.278 \cdot 10^{-6} \cdot mV^2\right) + \left(3.898 \cdot 10^{-3} \cdot mV\right) - 2.154\right) \cdot 100 \quad (2)$$

where.

%Pv = Volumetric water content, %.

mV = Soil moisture sensor output, mV.

Before the start of the irrigation season, the VWC at field capacity (FC: moisture retained in the soil against gravity) of each soil layer was determined. For this, the soil at 900 mm depth was saturated with water and allowed to seep into the lower soil layers under the influence of gravity. Sensor readings were taken during this time, and the value at which the sensor in each layer was fixed for 1 h was considered the soil moisture content at FC (%Pv) of the relevant layer (Table 10). The irrigation interval was 6–10 days during the irrigation season. At the end of each interval, the IWV to be applied was manually calculated using the difference between the sensor-measured actual soil moisture (SM) and FC determined by the user (Eq. 3). SM was considered as the average of two soil moisture sensors reading in each soil layer.

$$IWV = \sum_{i=0}^{2} (FC_i - SM_i) \cdot D_i \cdot A_i \quad (3)$$

where.

IWV = irrigation water volume, liter.

i = soil layer; in this case, three soil layers (0–300, 300–600, and 600–900 mm).

FC = soil moisture at field capacity of the soil layer i, %Pv.

SM = actual soil moisture (sensor measurement) of the layer i, %Pv.

D = depth of soil layer i, mm; 300 mm herein.

A = total irrigated area, m^2.

Total irrigated area (1,033.2 m^2) was calculated using Eq. 4.

$$A = \frac{D_t \cdot D_r \cdot S \cdot T}{100}, \quad (4)$$

where.

A = total irrigated area, m^2.

D_t = distance between two trees in the same row, 6 m.

D_s = distance between rows, 7 m.

S = shaded area per tree, 30%.

T = total number of trees, 82.

After manual calculations, the IWV was invoked by the "SWOPEN <liter water>" command (Table 9) by the user in a text window prepared on the web graphical user interface.

Climate and soil conditions

Monthly minimum and maximum temperatures (°C), relative humidity (%) and precipitation (mm) data covering the 2019, obtained from meteorological station established in Bornova-İzmir, are presented in Table 11. During the irrigation season (May-October) temperature and relative humidity ranged from 15.4 to 35.9 °C and 49.3 to 74.6%, respectively. A total of 85.4 mm of precipitation was recorded during this period. This meteorological data exhibits typical characteristics of Mediterranean climate zone.

Table 12 presents selected physical and chemical properties of the soil, including three distinct layers (0–200, 200–400 and 400–600 mm) from the test field. Based on texture classification, the soil is characterized as sandy loam. The measured pH values suggest a mildly alkaline environment. Electrical conductivity, determined from saturation paste measurements, indicates minimal salinity risk for crop production. The organic matter content across the soil layers ranged from 1.75% to 1.48%, reflecting a moderate level.

Field test results and discussion

During the irrigation season, the soil moisture (SM) varied between 42.0% and 24.4%, 44.6% and 25.9%, 46.7% and 30.2%, respectively, at soil depths of 0–300, 300–600, and 600–900 mm, based on the irrigation interval, FC

	0–300 mm	300–600 mm	600–900 mm
VWC at FC, %Pv	39.5	40.5	43.0

Table 10. Volumetric water content at the field capacity (FC) of olive orchard soil.

	Minimum temperature, °C	Maximum temperature, °C	Relative humidity, %	Precipitation, mm
January	5.4	12.7	83.1	369.3
February	5.2	15.4	75.6	106.3
March	7.2	19.9	65.4	37.8
April	10.9	22.5	61.9	66.1
May	15.4	28.2	61.1	12.6
June	21.5	33.8	56.3	21.7
July	22.3	34.7	51.7	8.8
August	24.4	35.9	49.3	1.3
September	18.9	30.7	60.3	18.4
October	15.5	26.6	74.6	22.6
November	12.1	23.3	79.8	58.2
December	7.7	16.1	78.1	73.4

Table 11. Monthly temperature, relative humidity and precipitation of Bornova-İzmir in 2019.

	Soil layer depth (mm)		
	0–200	200–400	400–600
Sand (%)	72	73	73
Silt (%)	12	12	12
Clay (%)	16	15	15
pH	7.6	7.5	7.6
EC (dS/m)	0.33	0.45	0.49
Organic matter (%)	1.75	1.52	1.48

Table 12. Some physical and chemical characteristics of olive orchard soil.

value, and applied IWV (Fig. 11). After each irrigation process, the SM content was lower in soil layers closer to the surface due to higher evapotranspiration.

Triggered by each irrigation, the flowrate increased up to 4–5 m^3/h in the first few minutes of irrigation because the irrigation pipes were filled with water and then stabilized around 2.8 m^3/h (Fig. 12), which corresponds to 0.23 m^3 throughout the 5-min irrigation data interval (Fig. 13). Due to irrigation events, a total irrigation water volume of 2,209 m^3 was supplied to 82 olive trees over the area of 1,033 m^2 during the irrigation season (Fig. 14), which is higher than the volumes recorded using LoRA and LoraWAN by Liopa-Tsakalidi et al[38]. as 4800, 2340 and 3600 m^3, respectively in 2019 (base year), 2020 and 2021, for 300 olive trees over 89,030 m^2 (22 acres). The differences observed in total water application can be attributed to variations in irrigation intervals (6–10 days vs 15–20 days), soil properties and climatic conditions. Due to the sandy-loam texture of the soil in this study, which is generally characterized by a higher infiltration rate and lower water-holding capacity, the more frequent irrigation schedule resulted in increased overall water use. However, Liopa-Tsakalidi et al.[38] did not provide details on soil characteristics. Additionally, differences in climatic conditions between the study sites could have influenced evapotranspiration rates, further impacting water requirements. In a study conducted at the same research institute by Aşık et al.[61] irrigation was applied at 25%, 50%, 75%, 100%, and 125% of Class-A pan evaporation, resulting in average seasonal water applications of 189, 377, 566, 754, and 943 mm over two years, respectively (tree spacing data not reported). Compared to the 100% pan evaporation treatment in that study, which received 754 mm of irrigation water, the total amount applied in the present study (2209 m^3 = 641 mm) was approximately 15% lower. This demonstrates that the proposed system achieved comparable irrigation performance with reduced water input.

The pipe inlet pressure was ~2 bar throughout the season and closely related to manually adjusted PRS-dial pressure regulator value (Fig. 15). The pressure peaked (~3.3 bar) at the beginning of July because the pressure regulator malfunctioned. Lower pressure values in Fig. 15 indicate insufficient water in the water resource. In drip irrigation, uniform water distribution throughout the field is negatively affected at a pressure lower than the optimal operating pressure of ~1 bar[62]. Moreover, the pipe inlet pressure of an irrigation plot is an excellent indicator of dripper clogging, pipe burst/leakage, or water availability in a water resource, i.e., wells and ponds, in any drip irrigation system. The pressure increases with excessive dripper clogging caused by poor irrigation water quality or ineffective filtration due to excessively high pressure, physical impact, or rodents.

Conclusion and future works

For the efficient use of water in agricultural irrigation, the irrigation system and soil properties must be effectively monitored and controlled. Drip irrigation systems and WSNs are viable solutions for improving agricultural water management techniques. Herein, WSN was used in an irrigation system for an olive orchard. By obtaining information on the SM content and IWV, users can make decisions by scheduling irrigations and reducing the

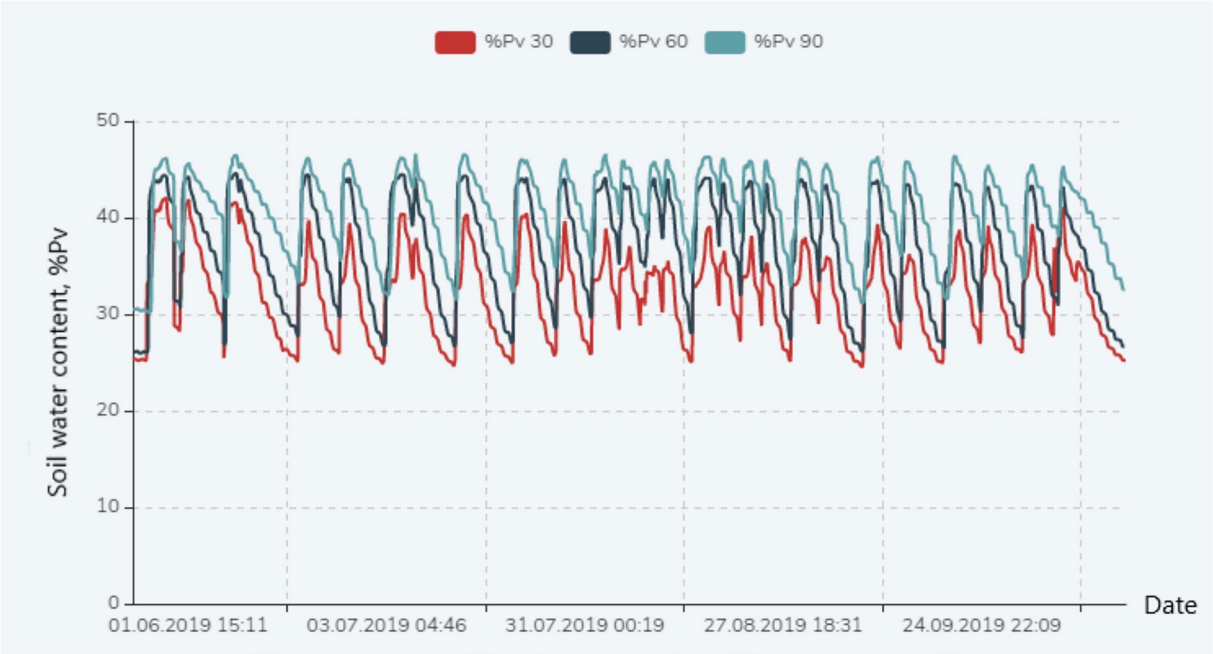


Fig. 11. Variation in the average soil water content in 0–300 (%Pv 30), 300–600 (%Pv 60), and 600–900 (%Pv 90)-mm soil layers in the soil moisture monitoring node.

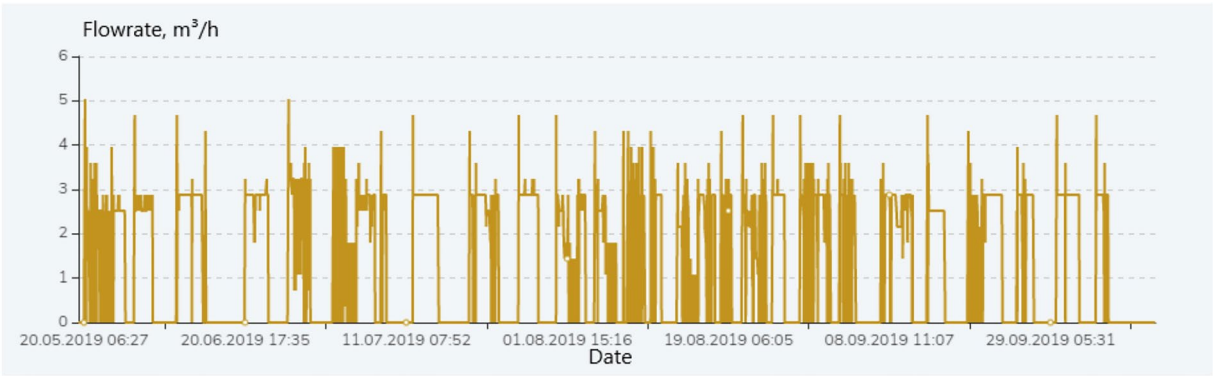


Fig. 12. Seasonal variation in the flowrate in the valve control node.

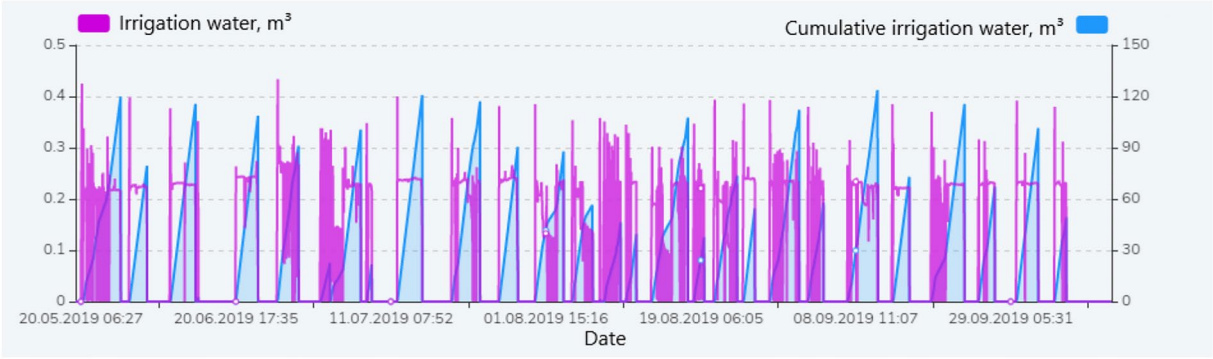


Fig. 13. Seasonal irrigation and cumulative irrigation water volume (IWV) during each irrigation in the valve control node.

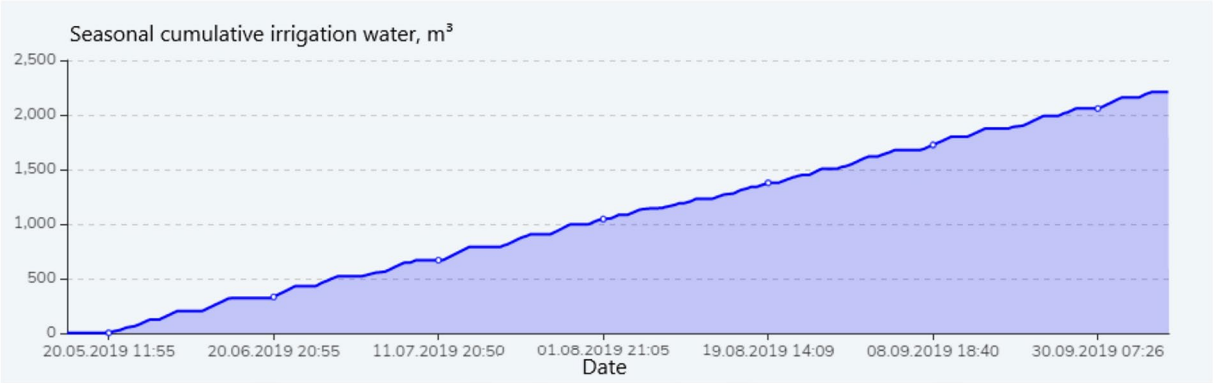


Fig. 14. Seasonal cumulative IWV in the valve control node.

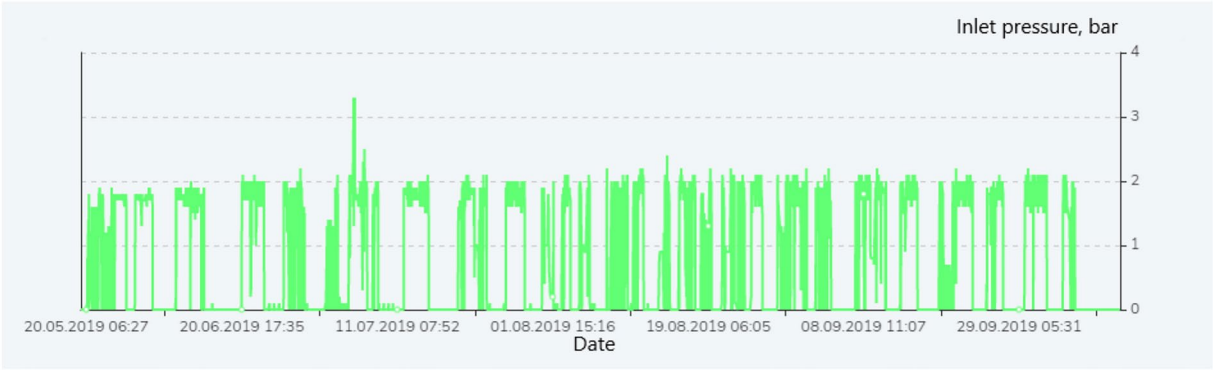


Fig. 15. Seasonal variation in the inlet pressure in the valve control node.

chance of over or under irrigation. WSNs also help the irrigation system to run efficiently and at peak efficiency by monitoring pressure and flowrate at the valve node, ensuring uniform water distribution. The findings from this study highlight the potential of integrating sensor-based monitoring and automation into irrigation practices, particularly in olive orchards, to achieve more sustainable water management.

This study introduced a custom-designed WSN architecture in terms of hardware, firmware and communication protocol, which differs significantly from prior research that primarily relied on commercial microcontroller platforms such as Arduino, Raspberry Pi, and ESP32. The proposed system was developed with a modular hardware structure consisting of main and submain layers, allowing for future modifications and integration of alternative communication modules such as LoRa and NB-IoT without altering the underlying sub-main hardware layer. In addition to the hardware modularity, a custom communication protocol was developed specifically for this system. Unlike standard communication stacks, the protocol was tailored to the specific data structure and readability.

The system's ability to monitor key irrigation parameters, including soil moisture content, total irrigation water volume, flowrate, and pipe inlet pressure, ensures precise irrigation control and early detection of anomalies such as dripper clogging and pressure fluctuations. Although pressure monitoring has not been widely implemented in irrigation systems, the field-tested pressure measurements in this study are consistent with other research demonstrating that pressure sensing[55–58], when integrated with wireless sensor networks and IoT technologies, can be effectively used for anomaly detection in water distribution networks.

Automated valve control further enhances irrigation efficiency by precisely controlling irrigation water volume, thereby minimizing human errors. The ability to monitor pressure variations in real time was another major contribution of this study, since the pressure measurement is a critical factor in maintaining optimal system performance. Compared to the conventional irrigation study by Aşık et al.[61] conducted in the same research institute, the proposed system achieved approximately 15% water saving. In contrast, when compared with the smart irrigation system developed by Liopa-Tsakalidi et al.[38], the total water applied in this study was relatively higher, which can be attributed to differences in climatic conditions, soil texture, and shorter irrigation intervals. Nevertheless, the findings indicate that further optimization of irrigation scheduling and parameter tuning may lead to additional water savings, potentially exceeding the 15% reduction already achieved.

In the study, the configuration structure of the firmware developed for both the soil monitoring and valve control nodes introduces a level of flexibility and functionality not commonly addressed in the current literature. Notably, the system allows for remote adjustment of key operational parameters, including the definition of

distinct data transmission intervals for irrigation and non-irrigation phases, which supports context-aware communication strategies. Additionally, the firmware enables the configuration of watermeter coefficients to accommodate the measurement of higher flowrates and cumulative water volumes, thereby increasing system adaptability to varying hydraulic conditions. Another significant feature is the independent scheduling of pressure control intervals, decoupled from data transmission cycles, which facilitates real-time system diagnostics. The firmware also supports a validation mechanism that allows the solenoid valve to close based on predefined thresholds of allowable water volume, followed by a pressure verification check to ensure complete valve closure. To the best of our knowledge, this integrated multi-parameter control logic, particularly the pressure verification step post valve actuation, has not been previously discussed or implemented in existing smart irrigation literature.

Although, the implications of this research extend beyond olive orchards to various agricultural enterprises as long-term agricultural operations depend on sustainable water management, several areas remain open for further improvement and research. Future work should focus on;

- Scaling the system for larger agricultural fields while maintaining energy efficiency and communication stability,
- Implementing the linear or polynomial calibration equation coefficients in the firmware, allowing different soil conditions to be accounted for and enabling remote modification of typical calibration equation to improve site-specific sensor accuracy,
- Optimizing the communication protocol by converting the current JSON-based data structure into a byte format, thereby reducing payload size and time-on-air duration, which in turn enhances energy efficiency and extends battery life in field deployments.
- Integrating LoRa and NB-IoT communication hardware and protocols, enabling flexible deployment in diverse agricultural environments,
- Conducting detailed energy consumption analysis by monitoring current and voltage levels of pumps and other electrical components using energy-meters or multimeters. This would enable the quantification of energy efficiency and allow comparison with conventional irrigation systems in terms of both energy and water use.
- Evaluating the use of AES-CTR encryption to ensure secure data transmission without increasing data payload size, thus preserving bandwidth efficiency while enhancing system security,
- Applying artificial intelligence algorithms and machine learning techniques, for instance implementing a dynamic decision mechanism that evaluates each incoming datasets both during irrigation and non-irrigation periods to determine and update the estimated start and end time of irrigation, for optimizing irrigation scheduling and water resource allocation.

Data availability

The datasets generated during and/or analysed during the current study are not publicly available due to privacy but are available from the corresponding author on reasonable request.

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M.K.M—Conceptualization, data curation, formal analysis, investigation, methodology, resources, software, visualization, writing – original draft, review & editing.

Declarations

Competing interests

The authors declare no competing interests.

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